

Jarvis March

1 Overview

Jarvis March

Jarvis March, also called the **gift-wrapping algorithm**, computes the convex hull by repeatedly choosing the next outermost point.

The algorithm behaves like wrapping a string around the outside of the point set:

- start from an extreme point;
- choose the next hull vertex;
- continue walking around the boundary;
- stop when the walk returns to the starting point.

Main Idea

At each hull vertex, choose the next point so that all other points lie on the same side of the directed edge.

2 Input and Output

Input:

$$S = \{P_1, P_2, \dots, P_n\}.$$

Output:

$$H = (H_1, H_2, \dots, H_h),$$

where H is the convex hull boundary and h is the number of hull vertices.

3 Algorithm Steps

1. Select the leftmost point P_0 . This means choose the point with minimum x -coordinate. If there is a tie, choose the point with minimum y -coordinate:

$$P_0 = \text{point with minimum } x, \text{ breaking ties by minimum } y.$$

This point is guaranteed to be on the convex hull because no point lies farther left than it.

2. Set current = P_0 .
3. Add current to the hull list.
4. Choose any point different from current as the temporary candidate.
5. Scan every other point R in the input set.

6. Compare R with the current candidate using

$$\text{orient}(\text{current}, \text{candidate}, R).$$

7. If R is more outside than the current candidate, replace the candidate:

$$\text{candidate} = R.$$

8. Continue scanning using the updated candidate.

9. After all points have been checked, the final candidate is the next hull vertex.

10. Set $\text{current} = \text{candidate}$ and repeat.

11. Stop when $\text{current} = P_0$, because the hull has wrapped back to the start.

Candidate Update Rule

For each hull point, Jarvis March does not choose the next vertex immediately. It keeps a temporary candidate, updates it while scanning all points, and chooses the final remaining candidate only after the full scan is finished.

4 Choosing the Next Vertex

Suppose the current hull point is P , the current candidate is Q , and another point being tested is R . Compute

$$\text{orient}(P, Q, R).$$

For

$$P = (P_x, P_y), \quad Q = (Q_x, Q_y), \quad R = (R_x, R_y),$$

the orientation value is

$$\text{orient}(P, Q, R) = (Q_x - P_x)(R_y - P_y) - (Q_y - P_y)(R_x - P_x).$$

Case	Decision
$\text{orient}(P, Q, R) < 0$	R is more clockwise than Q . Replace Q by R .
$\text{orient}(P, Q, R) > 0$	Keep Q .
$\text{orient}(P, Q, R) = 0$	Points are collinear. Keep the farther point from P .

Collinear Tie Rule

If Q and R are collinear from P , choose the point with larger squared distance:

$$d^2(P, R) > d^2(P, Q) \implies \text{choose } R.$$

5 Pseudocode

Jarvis March

```
JarvisMarch(points):  
    if number of points < 3:  
        return points  
  
    start = leftmost point  
    hull = empty list  
    current = start  
  
    repeat:  
        add current to hull  
        candidate = any point different from current  
  
        for each point r:  
            turn = orientation(current, candidate, r)  
  
            if r is more clockwise than candidate:  
                candidate = r  
  
            else if turn == 0 and r is farther from current:  
                candidate = r  
  
        current = candidate  
  
    until current == start  
  
    return hull
```

6 Worked Example

Problem

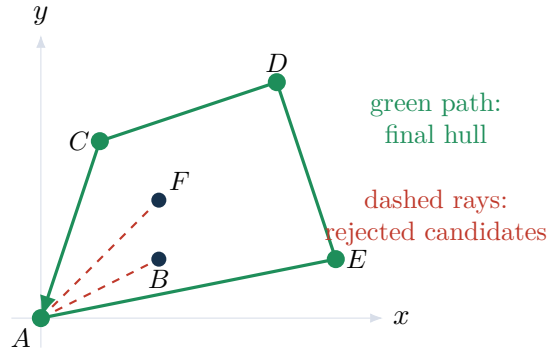
Apply Jarvis March to

$$\begin{aligned} A &= (0, 0), & B &= (2, 1), & C &= (1, 3), \\ D &= (4, 4), & E &= (5, 1), & F &= (2, 2). \end{aligned}$$

Step 1: Choose the Starting Point

The starting point is the leftmost point. If there is a tie, choose the lower one.

$\text{start} = A = (0, 0).$



Step 2: Find the Next Point from A

Set

$$\text{current} = A.$$

For this example, we wrap clockwise. Therefore:

$$\text{orient}(\text{current}, \text{candidate}, R) < 0 \implies R \text{ replaces the candidate.}$$

Start with B as a temporary candidate:

$$\text{candidate} = B.$$

The first useful replacement happens when E is tested. The vectors are

$$\overrightarrow{AB} = (2, 1), \quad \overrightarrow{AE} = (5, 1).$$

$$\text{orient}(A, B, E) = 2(1) - 1(5) = 2 - 5 = -3.$$

Since the value is negative, E is more clockwise than B . Replace the candidate:

$$\text{candidate} = E.$$

The full candidate trace from A is:

Test point	Orientation value	Candidate after test
C	$\text{orient}(A, B, C) = 5 > 0$	Keep B .
D	$\text{orient}(A, B, D) = 4 > 0$	Keep B .
E	$\text{orient}(A, B, E) = -3 < 0$	Replace by E .
F	$\text{orient}(A, E, F) = 8 > 0$	Keep E .

So the first hull edge is

$$A \rightarrow E.$$

The hull list is now

$$H = [A, E].$$

Step 3: Continue Wrapping

Repeat the same rule from the newest hull point. Each table shows how the temporary candidate changes while scanning the other points.

From E

Start with A as a temporary candidate.

Test point	Orientation value	Candidate after test
B	$\text{orient}(E, A, B) = -3 < 0$	Replace by B .
C	$\text{orient}(E, B, C) = -6 < 0$	Replace by C .
D	$\text{orient}(E, C, D) = -10 < 0$	Replace by D .
F	$\text{orient}(E, D, F) = 8 > 0$	Keep D .

So the next hull edge is

$$E \rightarrow D.$$

From D

Start with A as a temporary candidate.

Test point	Orientation value	Candidate after test
B	$\text{orient}(D, A, B) = 4 > 0$	Keep A .
C	$\text{orient}(D, A, C) = -8 < 0$	Replace by C .
E	$\text{orient}(D, C, E) = 10 > 0$	Keep C .
F	$\text{orient}(D, C, F) = 4 > 0$	Keep C .

So the next hull edge is

$$D \rightarrow C.$$

From C

Start with A as a temporary candidate.

Test point	Orientation value	Candidate after test
B	$\text{orient}(C, A, B) = 5 > 0$	Keep A .
D	$\text{orient}(C, A, D) = 8 > 0$	Keep A .
E	$\text{orient}(C, A, E) = 14 > 0$	Keep A .
F	$\text{orient}(C, A, F) = 4 > 0$	Keep A .

So the next hull edge is

$$C \rightarrow A.$$

The algorithm stops when the next point is the starting point A .

Worked Example Result

$$A \rightarrow E \rightarrow D \rightarrow C \rightarrow A$$

is the convex hull. The points B and F are not hull vertices.

7 Second Example: Few Hull Points

Problem

Find the hull of

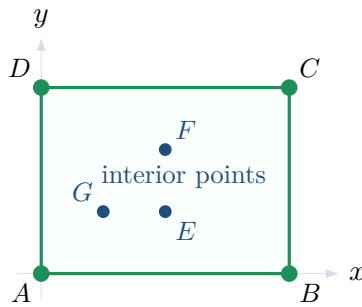
$$S = \{(0, 0), (4, 0), (4, 3), (0, 3), (2, 1), (2, 2), (1, 1)\}.$$

Name the points as

$$A = (0, 0), \quad B = (4, 0), \quad C = (4, 3), \quad D = (0, 3),$$

$$E = (2, 1), \quad F = (2, 2), \quad G = (1, 1).$$

The points E, F, G are inside the rectangle. Jarvis March still scans them, but they never remain as the final candidate.



Step 1: Start at A

The leftmost point is

$$A = (0, 0).$$

So A is the starting hull point.

Step 2: Choose the Next Point from A

Start with B as the temporary candidate. Now scan the remaining points.

Test point	Orientation value	Candidate after test
C	$\text{orient}(A, B, C) = 4(3) - 0(4) = 12 > 0$	Keep B .
D	$\text{orient}(A, B, D) = 4(3) - 0(0) = 12 > 0$	Keep B .
E	$\text{orient}(A, B, E) = 4(1) - 0(2) = 4 > 0$	Keep B .
F	$\text{orient}(A, B, F) = 4(2) - 0(2) = 8 > 0$	Keep B .
G	$\text{orient}(A, B, G) = 4(1) - 0(1) = 4 > 0$	Keep B .

After all points are checked, the final candidate is B . Therefore:

$$A \rightarrow B.$$

Step 3: Choose the Next Point from B

Start with A as the temporary candidate. Testing C gives a negative orientation, so C replaces A :

$$\text{orient}(B, A, C) = (-4)(3) - 0(0) = -12 < 0.$$

Now continue scanning using C as the candidate.

Test point	Orientation value	Candidate after test
D	$\text{orient}(B, C, D) = 0(3) - 3(-4) = 12 > 0$	Keep C .
E	$\text{orient}(B, C, E) = 0(1) - 3(-2) = 6 > 0$	Keep C .
F	$\text{orient}(B, C, F) = 0(2) - 3(-2) = 6 > 0$	Keep C .
G	$\text{orient}(B, C, G) = 0(1) - 3(-3) = 9 > 0$	Keep C .

After all points are checked, the final candidate is C . Therefore:

$$\boxed{B \rightarrow C.}$$

Step 4: Choose the Next Point from C

Start with A as the temporary candidate. Point B does not replace A :

$$\text{orient}(C, A, B) = (-4)(-3) - (-3)(0) = 12 > 0.$$

Testing D gives

$$\text{orient}(C, A, D) = (-4)(0) - (-3)(-4) = -12 < 0.$$

So D replaces A . Continue scanning using D as the candidate.

Testing the interior points gives:

$$\text{orient}(C, D, E) = (-4)(-2) - 0(-2) = 8 > 0,$$

$$\text{orient}(C, D, F) = (-4)(-1) - 0(-2) = 4 > 0,$$

$$\text{orient}(C, D, G) = (-4)(-2) - 0(-3) = 8 > 0.$$

All three values are positive, so E, F, G do not replace D .

After all points are checked, the final candidate is D . Therefore:

$$\boxed{C \rightarrow D.}$$

Step 5: Choose the Next Point from D

Start with A as the temporary candidate. Every remaining point stays on the left side of $D \rightarrow A$, so A remains the candidate.

Test point	Orientation value	Candidate after test
B	$\text{orient}(D, A, B) = 0(-3) - (-3)(4) = 12 > 0$	Keep A .
C	$\text{orient}(D, A, C) = 0(0) - (-3)(4) = 12 > 0$	Keep A .
E	$\text{orient}(D, A, E) = 0(-2) - (-3)(2) = 6 > 0$	Keep A .
F	$\text{orient}(D, A, F) = 0(-1) - (-3)(2) = 6 > 0$	Keep A .
G	$\text{orient}(D, A, G) = 0(-2) - (-3)(1) = 3 > 0$	Keep A .

The next point is the starting point A , so the algorithm stops.

Second Example Result

$$A \rightarrow B \rightarrow C \rightarrow D \rightarrow A$$

That is,

$$(0, 0) \rightarrow (4, 0) \rightarrow (4, 3) \rightarrow (0, 3) \rightarrow (0, 0)$$

is the convex hull. The interior points E, F, G are checked during the scan, but none of them remains as the final candidate.

8 Complexity

Let:

$$n = \text{number of input points}, \quad h = \text{number of hull vertices}.$$

For each hull vertex, Jarvis March scans all n points to find the next hull vertex. Therefore:

$$\text{time complexity} = \mathcal{O}(nh).$$

The algorithm is output-sensitive because the running time depends on h , the number of hull vertices.

When Jarvis March Is Good

Jarvis March is especially useful when the number of hull vertices h is small compared with the total number of points n .

9 Graham Scan vs Jarvis March

Feature	Graham Scan	Jarvis March
Main idea	Sort by angle, then use a stack.	Wrap around the outside.
Starting point	Lowest point.	Leftmost point.
Key operation	Orientation while scanning stack.	Orientation while choosing next hull point.
Time complexity	$\mathcal{O}(n \log n)$.	$\mathcal{O}(nh)$.
Best when	General use, many hull points.	Few hull points.

10 Final Review

Jarvis March Summary

Start at the leftmost point.
Repeatedly choose the next outermost point.
Use orientation to compare candidates.
For collinear candidates, choose the farther point.
Stop when the start point is reached again.
Running time is $\mathcal{O}(nh)$.